

Reviewer Pack — Minimal Order of Quaternionic Kähler Forms and Circuit Monot...

Entry 1b9dc3c0e047 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Quaternionic Kähler Forms and Circuit Monotone Width Inequality

Entry ID: 1b9dc3c0e047 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-04 04:50:10 UTC

1. Conjecture

Field A (mathematical branch): Quaternionic Geometry (Kähler Forms) **Field B** (complexity object): Boolean Circuit Complexity (Circuit Monotone Width)

Statement:

For every n -variable Boolean circuit C with monotone width $w(C)$, there exists a quaternionic Kähler form Q such that the minimal order of its associated integral model is $O(w(C)^2)$. Equivalently, for all circuits C , $|O(Q)| \leq \Theta(w(C)^2)$, where $O(Q)$ denotes the set of orders of the quaternionic Kähler form.

Rationale (proposer's reasoning):

Quaternionic Kähler forms provide a rich algebraic structure that may encode geometric properties of complex varieties. By studying their minimal order in relation to circuit monotone width, we might uncover new insights into the complexity of Boolean circuits and potentially expose novel connections between algebraic geometry and computational complexity.

Taxonomy category: Quaternionic_Kahler_Forms_to_Circuit_Monotone_Width (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 6a322a901f065fc3

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a given Boolean circuit C with monotone width $w(C)$, $|O(Q)| \leq \Theta(w(C)^2)$ is supported if, across 30 random seeds, the mean of $|O(Q)|$ values from the associated quaternionic Kähler forms Q are within ± 3 standard deviations of $O(w(C)^2)$. Falsified if any seed has an $|O(Q)|$ value exceeding $O(w(C)^2)$ by more than 10 units or if $\text{support_fraction} < 0.8$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.2s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_monotone_circuit(n, w):
        circuit = []
        for _ in range(w):
            gate = random.choice(['AND', 'OR'])
            inputs = [random.randint(0, 1) for _ in range(n)]
            circuit.append((gate, inputs))
        return circuit

    def compute_minimal_order(circuit):
        order = 0
        for gate, inputs in circuit:
            if gate == 'AND':
                order += sum(inputs)
            elif gate == 'OR':
                order += max(inputs)
        return order

    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        for _ in range(5):
            circuit = generate_monotone_circuit(n, random.randint(1, min(40, n)))
            order = compute_minimal_order(circuit)
            results.append({"n": n, "order": order})
```

```

mean_order = sum(result["order"] for result in results) / len(results)
std_dev = math.sqrt(sum((result["order"] - mean_order) ** 2 for result in results) / len(results))

conjecture_holds = all(abs(order - (n // 2) ** 2) <= 10 for n, order in zip(n_values, [result["order"] for result in results]))
counterexample = "" if conjecture_holds else f"order={max([result['order'] for result in results])}"

return {
    "metric_name": "minimal_order",
    "metric_value": mean_order,
    "instances_tested": len(results),
    "n_max": max(n_values),
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(result["metric_value"] for result in results) / len(results)
    std_dev_metric_value = math.sqrt(sum((result["metric_value"] - mean_metric_value) ** 2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_dev_metric_value} support_fraction={support_fraction}")
    elif any(abs(result["metric_value"] - (n // 2) ** 2) > 10 for n, result in zip(n_values, [result["order"] for result in results])):
        first_failing_seed = next(seed for seed, result in enumerate(results, start=1) if abs(result["metric_value"] - (n // 2) ** 2) > 10)
        print(f"RESULT: FALSIFIED counterexample='order exceeds expected 0(w(C)^2)' first_failing_seed={first_failing_seed}")
    else:
        print(f"RESULT: INCONCLUSIVE support_fraction={support_fraction}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

, 'counterexample': 'order=276 exceeds expected 0(w(C)^2)'}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 67.53333333333333, 'instances_tested': 30, 'n_max': 40}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 79.46666666666667, 'instances_tested': 30, 'n_max': 40}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 77.33333333333333, 'instances_tested': 30, 'n_max': 40}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 80.63333333333334, 'instances_tested': 30, 'n_max': 40}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 100.53333333333333, 'instances_tested': 30, 'n_max': 40}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 75.96666666666667, 'instances_tested': 30, 'n_max': 40}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 67.03333333333333, 'instances_tested': 30, 'n_max': 40}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 77.1, 'instances_tested': 30, 'n_max': 40}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 72.16666666666667, 'instances_tested': 30, 'n_max': 40}

```

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f140e010.py", line 77, in <module>

```
elif any(abs(result["metric_valu
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge falsified | original: The test results show that for multiple trials, the minimal order of the quaternionic Kähler forms exceeds the expected value $O(w(C)^2)$ by more than 1 | next: Investigate further to understand why the minimal orders are exceeding the expected values and explore alternative methods or theories that could explain this discrepancy.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	17930
2	preregistration	ollama_remote	glm4:latest	0	0	10192
3	novelty	ollama_remote	glm4:latest	0	0	17267
4	novelty	ollama_remote	glm4:latest	0	0	15834
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16936
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	22561
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9921
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10629
9	judge	ollama_remote	glm4:latest	0	0	9277

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 130547 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/1b9dc3c0e047.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/1b9dc3c0e047.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/1b9dc3c0e047.tar.gz` (if generated)